Explanations Go Linear: Post-hoc Explainability for Tabular Data with Interpretable Meta-Encoding

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APPENDIX

A. Detailed Experimental Settings

Datasets. We implement synthetic classifiers from SENECA framework [1] by varying the dimensions of the synthetic instances. In particular, we adjust the number of informative features (t), and uninformative features (u). The total number of features, $m = t + u$, is set in the list $\{4, 8, 16, 32, 64\}$ and, for a fixed $t+u$, we define $t = \min\{16, t+u\}$. For each dataset dimension, we generate five rule-based classifiers and five linear classifiers. Hence, we explain 2,048 instances for each of them. Besides synthetic data, we employ 18 real-world datasets from UCI ML Repo[1]. In every dataset, we apply standard scaling, with zero mean and unitary variance to continuous features, and one-hot encoding to categorical features.

Black-boxes. As black-box classifiers, we consider the ensemble methods XGBoost (XGB) [2], LightGBM (LGB) [3] and CatBoost (CTB) [4], because they are among the most effective techniques for tabular data, even outperforming deep learning models [5], [6]. Anyway, there is no specified limitation in using other black-boxes, such as NNs or SVMs. Each black-box is trained using 80% of the dataset, with the remaining 20% reserved for testing and evaluation. The black-box models are trained on each dataset's classification tasks, whether binary or multi-class. For binary datasets, each explanation method is asked to generate explanations for the class 1. For multi-class datasets, each explanation method is asked to generate explanations for the majority class.

Experimental setup for ILLUME. The function f^b is implemented as a 3-layer fully-connected neural network. We did not tune the architecture with respect to the best number of layers and/or hidden sizes. Meta-encoder training is done with Adam optimizer, fixing learning rate to 10^{-3}, with early stopping technique to prevent overfitting. To compute the loss functions, for the latent space vectors we use the cosine distance[2], $d_{\mathcal{Z}}(i, j) = d_{cos}(\mathbf{z}_i, \mathbf{z}_j)$. In line with [7]–[9], we divide the input distance into contributions from continuous and categorical (one-hot encoded) features. Assuming h categorical features after one-hot encoding, we express the input distance[3] as $d_{\mathcal{X}}(i, j) = \frac{h-m}{m} d_{cos}(\mathbf{x}_i^{con}, \mathbf{x}_j^{con}) + \frac{h}{m} d_{hmm}(\mathbf{x}_i^{cat}, \mathbf{x}_j^{cat})$. For the black-box score vectors, we employ cosine distance as well. For latent space transformation matrices, we consider the average per-column cosine distance $d_{\mathcal{W}}(i, j) = \frac{1}{k} \sum_r d_{cos}(W^b_{i,:,r}, W^b_{j,:,r})$. All the encodings are tested tuning the latent space dimension as hyperparameter from the list $\{2, 4, 8, 16, 32\}$.

[1]https://archive.ics.uci.edu/ml/index.php
[2]$d_{cos}(\mathbf{u}, \mathbf{v}) = 1 - \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||\ ||\mathbf{v}||}$
[3]$d_{hmm}(\mathbf{u}, \mathbf{v}) = \frac{1}{len(\mathbf{u})} \sum_i \mathbb{1}[u_i \neq v_i]$

Experimental Setup for Dimensionality Reduction. We compare ILLUME against latent embedding methods to evaluate the neighborhood structure preservation in the latent space. Considered methods for dimensionality reduction are PCA[4], ISOMAP[5], LLE[6] and p-UMAP[7]. As for ILLUME, the reduction methods are tested tuning the latent space dimension as hyperparameter from the list $\{2, 4, 8, 16, 32\}$. For p-UMAP, we train a 3-layer fully-connected neural network for 50 epochs. For the other methods, we used standard parameters.

Experimental Setup for Local Explainers. We compare ILLUME with well-known local explainers to evaluate the quality of explanations: LIME[8], SHAP[9] for feature importance; LORE[10], ANCHOR[11] for decision rules. LIME is tuned with neighborhood sizes $\{100, 300, 1000, 5000\}$; LORE with sizes $\{300, 1000\}$ and $\{1, 5, 10\}$ decision trees; ANCHOR with batch sizes $\{100, 300\}$ and beam sizes $\{4, 10\}$. With SHAP we use the kernelSHAP method for synthetic black-boxes and treeSHAP for ensemble-tree black-boxes.

Experimental Setup for Global Surrogates. Global surrogates models LR[12] and DT[13] are trained with latent or input space variables by tuning their main hyperparameters to maximize test set prediction accuracy. For surrogates based on LR (INP-LR and LIN-LR), as local explanation the –global– logistic coefficients $\{\beta_1, \beta_2, \dots\}$ are weighted by the feature values of the specific instance (see model-intrinsic additive scores [10]), namely $\psi^{\text{INP}}_{i,j} = \beta_j x_{i,j}$ and $\psi^{\text{LIN}}_{i,j} = \sum_{r=1}^{k} \beta_r W^b_{j,r} x_{i,j}$. For a fair comparison, we apply the latent search for maximizing surrogate fidelity also for LIN-LR and LIN-DT using distance $d_{\mathcal{Z}}$. Instead, for LR and DT we search the nearest neighbor in the input space according to $d_{\mathcal{X}}$. For remaining methods –LIME, SHAP, LORE and ANCHOR– we trust each explanation as valid one, since all these methods ensure guarantees for maximal fidelity.

[4]https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html
[5]https://scikit-learn.org/stable/modules/generated/sklearn.manifold.Isomap.html
[6]https://scikit-learn.org/stable/modules/generated/sklearn.manifold.LocallyLinearEmbedding.html
[7]https://umap-learn.readthedocs.io/en/latest/parametric_umap.html
[8]https://lime-ml.readthedocs.io/en/latest/lime.html\#module-lime.lime_tabular
[9]https://shap.readthedocs.io/en/latest/generated/shap.Explainer.html
[10]https://kdd-lab.github.io/LORE_sa/html/index.html
[11]https://github.com/marcotcr/anchor
[12]https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html
[13]https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html

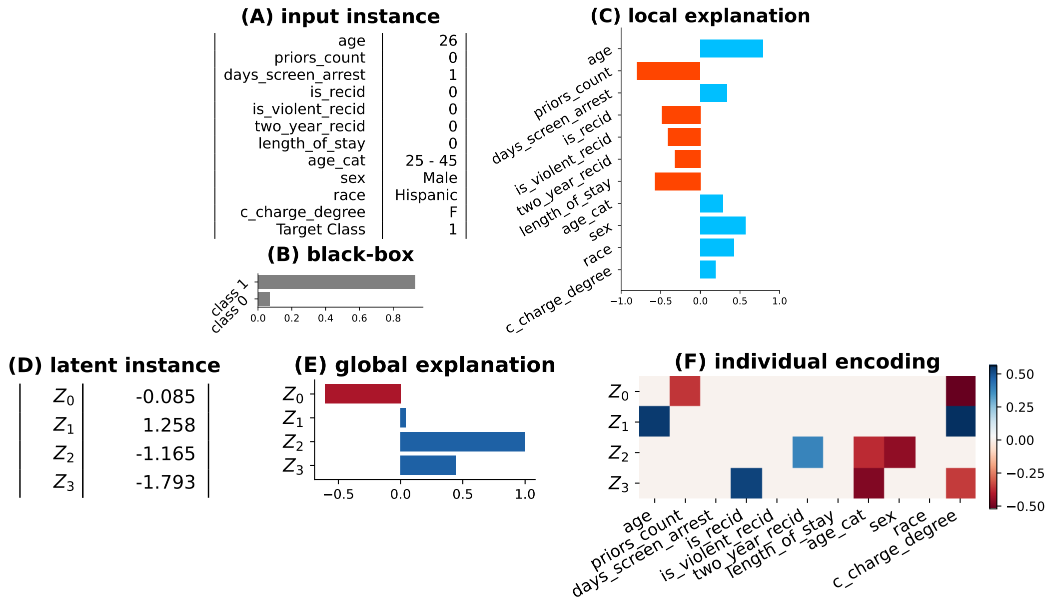

Fig. A1: Exemplification of ILLUME's inference phase on `compas` dataset when generating feature importance local explanations. Given an input data-point $\mathbf{x}_{test}$ (A) and its corresponding black-box prediction $b(\mathbf{x}_{test})$ (B), the method outputs instance-specific explanation $e_g(\mathbf{z}_{test}, \eta^b_{test})$ (C). This explanation is derived: *(i)* by encoding the instance into a latent representation $\mathbf{z}_{test}$ (D), *(ii)* extracting the logic of the global surrogate g (E), and *(iii)* combining it with local interpretable mapping η^b_{test} (F), represented by a sparse and linear transformation returned by the meta-encoder.

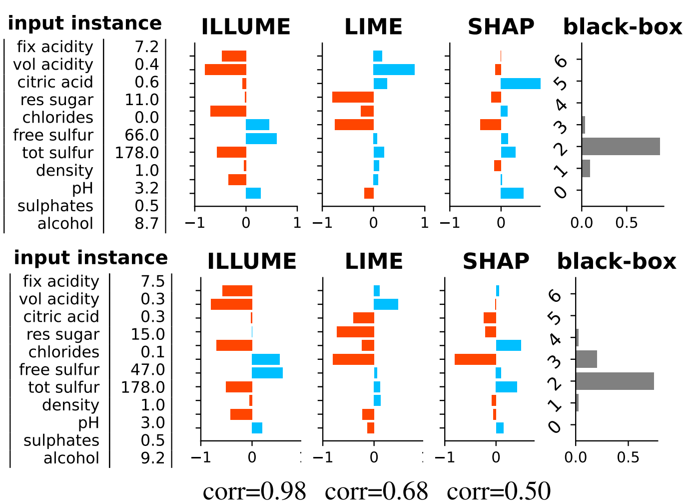

Fig. A2: Feature importance explanations for two similar records of the `wine` dataset. Feature values are reported on the left. In the center, explanations are derived with ILLUME, LIME and SHAP. On the right, the prediction probability returned by an XGB classifier.

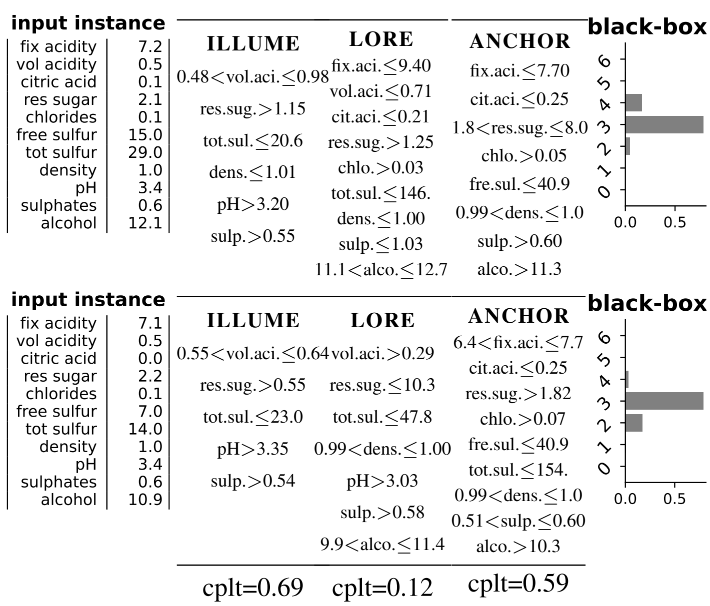

Fig. A3: Decision rule explanations for two similar records of the `wine` dataset. Feature values are reported on the left. In the center, explanations derived with ILLUME, LORE and ANCHOR. On the right, the prediction probability returned by an LGB classifier.

B. Qualitative Results

In Figure A1 it is illustrated how our approach works in the explanation inference phase with an example on `compas` dataset. At first glance, with ILLUME, instances and their local black-box decisions (A-B) are employed to generate local explanations (C). These explanations are computed with the information deriving from an auxiliary embedding space where we project the input (D). In this space we have extracted global latent explanations (E) during the surrogate learning process. The key component of this space is a set of instance-specific linear transformations (F) learned during the meta-encoder training process. Local explanations are produced by combining the latent explanation with the local projection of the data. Figure A2 showcase feature importance explanations for two neighboring instances from the `wine` dataset. Being closely located in the feature space and yielding similar predictions from the black-box model for the multi-class task, the explanations for these instances are expected to align, ranking important features in comparable ways. ILLUME demonstrates its ability to generate robust and consistent explanations, producing rank-preserving feature importance profiles for neighboring instances. In contrast, competing methods such as LIME and SHAP fail to achieve similar consistency. Additionally, Figure A3 presents a comparable example for decision rules using the same dataset, illustrating that ILLUME is capable of generating robust and consinstent rules. The similarity of these rules is evaluated using the *cplt-score*, like in the main experimental result of the paper. Furthermore, we demonstrate that ILLUME can produce more concise decision rules with respect to the analyzed competitors, a benefit achieved through sparsity regularization.

C. Detailed Formulas for Decision Rules

In ILLUME, we first derive global explanations in the form of axis-parallel decision rules on latent features, determined by root-leaf paths in a trained decision tree, i.e., $\rho_i^z = \{z_{i,r} \in [l_r^z, u_r^z]\}_{r=1\dots k}$ (lower and upper bound are in the domain of $z_{i,r}$, extended with $\pm\infty$). By exploiting the linearity of the mapping η_i^b, these rules are converted into input space local oblique rules $\tilde{\rho}_i^x = \{\sum_{j=1}^m W_{i,j,r}^b x_{i,j} \in [l_r^z, u_r^z]\}_{r=1\dots k}$. While oblique rules offer a valid and expressive form of explanation, for consistency with most methods we perform an additional step to convert the oblique rules into the standard axis-parallel format: $\rho_i^x = \{x_{i,j} \in [l_{i,j}^x, u_{i,j}^x]\}_{j=1\dots m}$. The upper and lower bounds for these axis-parallel rules are derived by isolating $x_{i,j}$ to express the rule in terms of individual input features. For example, from the oblique rules $\sum_{j=1}^m W_{i,j,r}^b x_{i,j} \geq l_r^z$ and $\sum_{j=1}^m W_{i,j,r}^b x_{i,j} \leq u_r^z$, it follows:

$$l_{i,j}^x = \max_r \frac{l_r^z - \sum_{v\neq j} W_{i,v,r}^b x_{i,v}}{W_{i,j,r}^b}$$

$$u_{i,j}^x = \min_r \frac{u_r^z - \sum_{v\neq j} W_{i,v,r}^b x_{i,v}}{W_{i,j,r}^b}.$$

The max/min operations ensure taking the most restrictive inequality among the k latent conditions, i.e., the largest lower bound and the smallest upper bound. Finally, exploiting the equality $z_{i,r} = \sum_{v\neq j} W_{i,v,r}^b x_{i,v} + W_{i,j,r}^b x_{i,j}$, we find the relations reported in the main paper that link upper/lower bounds in the embedding and in the original space:

$$l_{i,j}^x - x_{i,j} = \max_r \frac{l_r^z - z_{i,r}}{W_{i,j,r}^b} \qquad u_{i,j}^x - x_{i,j} = \min_r \frac{u_r^z - z_{i,r}}{W_{i,j,r}^b}.$$

D. Perfect Fidelity via Similarity Search

Relying on surrogate logic, in ILLUME the explanations $e_g(\mathbf{z}_i, \eta_i^b)$ are valid iff $g(\mathbf{z}_i) = b(\mathbf{x}_i)$, i.e., when surrogate model agrees with the black-box on the corresponding instance. To ensure producing valid explanations for *every* instance, we refine the latent representation of those samples for which $g(\mathbf{z}_i) \neq b(\mathbf{x}_i)$. The goal of this refinement is to realign misclassified latent points with nearby correctly predicted ones that share the same black-box label. This approach leverages ILLUME's design, which ensures that nearby points in latent space have similar transformations. Hence, explanations for these points remain closely aligned and reliable. Thus, for each $\mathbf{z}_i \in Z$ such that $g(\mathbf{z}_i) \neq b(\mathbf{x}_i)$, we proceed as follows:

(1) Closest valid neighbor search. We perform a cosine-based nearest-neighbor search in latent space to identify $\mathbf{z}_{nn} \in Z$ such that $g(\mathbf{z}_{nn}) = b(\mathbf{x}_{nn})$ and $b(\mathbf{x}_i) = b(\mathbf{x}_{nn})$. This provides a nearby latent point that is locally consistent with the black-box prediction and serves as a reference.

(2) First-order approximation. Since $\mathbf{z}_{nn}$ is the closest latent vector to $\mathbf{z}_i$, we assume their corresponding inputs differ by a small displacement (i.e., $\mathbf{x}_{nn} \approx \mathbf{x}_i + \delta$). Similarly, we assume the associated local transformations satisfy $W_{nn}^b \approx W_i^b + \Delta W_i$. We use a first-order approximation from the formula $W_{nn}^b \mathbf{x}_{nn} = (W_i^b + \Delta W_i)(\mathbf{x}_i + \delta)$ to obtain:

$$\mathbf{z}_{nn} \approx \mathbf{z}_i + J_i\delta = W_i^b \mathbf{x}_i + \Delta W_i \mathbf{x}_i + W_{nn}^b \delta + \mathcal{O}(||\Delta W_i \delta||).$$

This decomposition separates the effect of input displacement δ with the projection variation ΔW_i. Here, $W_{nn}^b \delta \approx W_i^b \delta$ because we neglected second-order terms.

(3) Interpolated latent representation. Guided by the previous approximation, we define the latent perturbation:

$$\mathbf{z}_i^\gamma = W_i^b \mathbf{x}_i + \gamma_W (W_{nn}^b - W_i^b)\mathbf{x}_i + \gamma_x W_{nn}^b (\mathbf{x}_{nn} - \mathbf{x}_i),$$

where $(W_{nn}^b - W_i^b)$ and $(\mathbf{x}_{nn} - \mathbf{x}_i)$, both scaled by small factors γ_W and γ_x, identify the directions of the perturbations δ and ΔW_i. This expression can be interpreted as an interpolation toward the nearest valid neighbor:

$$\mathbf{z}_i^\gamma = (1 - \gamma_W)\mathbf{z}_i + \gamma_W W_{nn}^b \mathbf{x}_i + \gamma_x W_{nn}^b (\mathbf{x}_{nn} - \mathbf{x}_i).$$

(4) Minimal constrained perturbation. We determine the optimal interpolation parameters by solving

$$\gamma_W^*, \gamma_x^* = \underset{(\gamma_W,\gamma_x)\in(0,1]^2}{\arg\min} ||\mathbf{z}_i^\gamma - W_i^b \mathbf{x}_i||^2 \text{ s.t. } g(\mathbf{z}_i^\gamma) = b(\mathbf{x}_i).$$

This ensures that the refined latent vector remains as close as possible to the original one, while restoring the agreement between surrogate and black-box predictions. Notably, at least the solution $(\gamma_W^* = 1, \gamma_x^* = 1)$ exists because $\mathbf{z}_i^{(\gamma_W=1,\gamma_x=1)} \equiv \mathbf{z}_{nn}$ and $g(\mathbf{z}_{nn}) \equiv b(\mathbf{x}_i)$.

The refined latent instance is then given by $\mathbf{z}_i^* = W_i^* \mathbf{x}_i + \boldsymbol{\varepsilon}_i^*$, where $W_i^* = W_i^b + \gamma_W^*(W_{nn}^b - W_i^b) \equiv W_i^b + \Delta W_i$ is a updated transformation matrix and $\boldsymbol{\varepsilon}_i^* = \gamma_x^* W_{nn}^b (\mathbf{x}_{nn} - \mathbf{x}_i) \equiv W_{nn}^b \delta$ is an induced offset. In practice, we approximate the solution via grid-search over $(0,1]^2$. Among feasible solutions, we select the one minimizing $\gamma_W + \gamma_x$, favoring minimal corrections in both transformation and input space. Overall, the refinement corresponds to the smallest local perturbation of the latent mapping that restores surrogate–black-box agreement.

This approach enables the construction of a valid explanation even for those instances where the surrogate is not capable to replicate black-box label. Specifically, for feature importance explanations, the attribution vectors will be linearly updated:

$$\psi^*_{i,j} = \sum_{r=1}^{k} \beta_r (W^b_{i,j,r} + \Delta W_{i,j,r}) \equiv \psi_{i,j} + \Delta \psi_{i,j}$$

Furthermore, for decision rule explanations, we will have corrected oblique rules with translated upper/lower bounds as:

$$\tilde{\rho}^*_i = \Big\{ \sum_{j=1}^{m} (W^b_{i,j,r} + \Delta W_{i,j,r}) x_{i,j} \in [l^z_r - \varepsilon^*_{i,r}, u^z_r - \varepsilon^*_{i,r}] \Big\}_{r=1\ldots k}$$

Once converted to axis-aligned rules $\rho^*_i = \{x_{i,j} \in [l^*_{i,j}, u^*_{i,j}]\}_{j=1\ldots m}$, they will satisfy the following:

$$l^*_{i,j} - x_{i,j} = \max_r \frac{l^z_r - z^*_{i,r}}{W^b_{i,j,r} + \Delta W_{i,j,r}}$$

$$u^*_{i,j} - x_{i,j} = \min_r \frac{u^z_r - z^*_{i,r}}{W^b_{i,j,r} + \Delta W_{i,j,r}}.$$

E. Counterfactual Rules and Counter-Examples

While decision rules are directly obtained from the root-to-leaf paths of a decision tree, counterfactual rules are derived through symbolic reasoning applied to the same tree. Following the methodology proposed in [7], we analyze the latent decision tree (already used to generate the decision rules) to identify paths leading to a prediction opposite to that of the input instance.

For each test instance, we first rank training instances whose surrogate tree predictions differ from that of the test instance based on the cosine similarity between their latent representations. We then examine the decision rules associated with these instances and select those requiring the minimal number of split condition changes with respect to the rule satisfied by the test instance. This strategy identifies the closest instances that achieve the desired prediction change with the smallest possible modifications, thereby optimizing both proximity and sparsity, which are key properties of high-quality counterfactual explanations [11].

Finally, we also return the corresponding training instances from which the counterfactual rules are derived, providing them as counterfactual examples.

F. Analysis of Stability Loss

Here, we provide an intuition behind the stability loss computation, which guarantees that small changes in the original data space lead to minimal perturbations in the latent space, thus ensuring a consistent and reliable mapping across data points. We enforce each local transformation to remain valid when applied to slightly perturbed input. Specifically, omitting b for simplicity, if the encoding is given by $\mathbf{z} = \eta(\mathbf{x}) = f(\mathbf{x}) \cdot \mathbf{x}$, where $f(\mathbf{x})$ represents the individual transformation for $\mathbf{x}$, then the same transformation must hold for a small perturbation $\mathbf{x} + \delta$. In other words, the encoding operates locally around the instance $\mathbf{x}$ as a linear transformation characterized by a matrix $f(\mathbf{x})$, which is approximately constant within the neighborhood of $\mathbf{x}$. While the coefficients of the matrix dynamically adapt to the input, their rate of variation is slower than that of the input $\mathbf{x}$, ensuring stability and consistency of the transformation. This implies the following constraint:

$$\eta(\mathbf{x} + \delta) = f(\mathbf{x} + \delta) \cdot (\mathbf{x} + \delta) \approx f(\mathbf{x}) \cdot (\mathbf{x} + \delta).$$

We enforce this property by minimizing specific quantities which ensure the validity of the equation above. First, we show the first-order Taylor expansions of matrix entries of the local transformation f under small input perturbations:

$$f_{j,r}(\mathbf{x} + \delta) = f_{j,r}(\mathbf{x}) + \sum_{v=1}^{m} \frac{\partial f_{j,r}(\mathbf{x})}{\partial x_v} \delta_v + \mathcal{O}(||\delta||^2)$$

Substituting these approximations into the expression for the encoding of the perturbation, $\eta(\mathbf{x} + \delta) = f(\mathbf{x} + \delta) \cdot (\mathbf{x} + \delta)$, for each entry of the vector we get (neglecting second-order terms):

$$\begin{aligned}
\eta_r(\mathbf{x} + \delta) &\approx \sum_{j=1}^{m} \Big(f_{j,r}(\mathbf{x}) + \sum_{v=1}^{m} \frac{\partial f_{j,r}(\mathbf{x})}{\partial x_v} \delta_v \Big) (x_j + \delta_j) \\
&= \sum_{j=1}^{m} f_{j,r}(\mathbf{x})(x_j + \delta_j) + \\
&+ \sum_{j=1}^{m} \Big(\sum_{v=1}^{m} \frac{\partial f_{v,r}(\mathbf{x})}{\partial x_j} x_v \Big) \delta_j + \mathcal{O}(||\delta||^2).
\end{aligned}$$

Last equation tells us that describing the variation of the latent encoding η, when applied to a minimal perturbation of $\mathbf{x}$, requires the sum of two quantities (up to second-order corrections):

$$\eta(\mathbf{x}+\delta) \approx f(\mathbf{x})(\mathbf{x}+\delta) + D(\mathbf{x}) \cdot \delta \qquad \Big[D_{j,r} = \sum_{v} \frac{\partial f_{v,r}}{\partial x_j} x_v \Big]$$

The first term, $f(\mathbf{x}) \cdot (\mathbf{x} + \delta)$, represents the mapping of the perturbation δ applied to the input $\mathbf{x}$, with the linear transformation f held constant. The second term, $D(\mathbf{x}) \cdot \delta$, captures the change in the transformation due to the perturbation δ and its interaction with the input $\mathbf{x}$. Therefore, we enforce stability by minimizing $||D(\mathbf{x})||^2_{\mathrm{F}}$ during training. Reordering the terms $\eta(\mathbf{x} + \delta) \approx f(\mathbf{x}) \cdot \mathbf{x} + (f(\mathbf{x}) + D(\mathbf{x})) \cdot \delta$, we obtain the Taylor expansion of the encoding η under small input perturbations:

$$\eta(\mathbf{x} + \delta) \approx \eta(\mathbf{x}) + J(\mathbf{x}) \cdot \delta \qquad \Big[J(\mathbf{x}) = f(\mathbf{x}) + D(\mathbf{x}) \Big]$$

where we highlight the Jacobian matrix J around the data-point $\mathbf{x}$, with entries $J_{j,r} = \frac{\partial \eta_r}{\partial x_j}$. To minimize $||D(\mathbf{x})||^2_{\mathrm{F}}$, we optimize the reported loss involving Jacobian matrix $L^{st}(\mathbf{x}) = ||J(\mathbf{x}) - f(\mathbf{x})||^2_{\mathrm{F}}$ for each instance.

G. Detailed Experimental Results

Here, we report in details all the results that are shown in aggregated form in the main paper.

RQ1 - Goodness of Latent Space Quality

In Tables A1, A4 and A5 are reported extensive results for all metrics, datasets and black-boxes that have been summarized in the main paper to answer **RQ1**. In particular, central and right columns of Tables A4 and A5 report Macro-F1 accuracies for

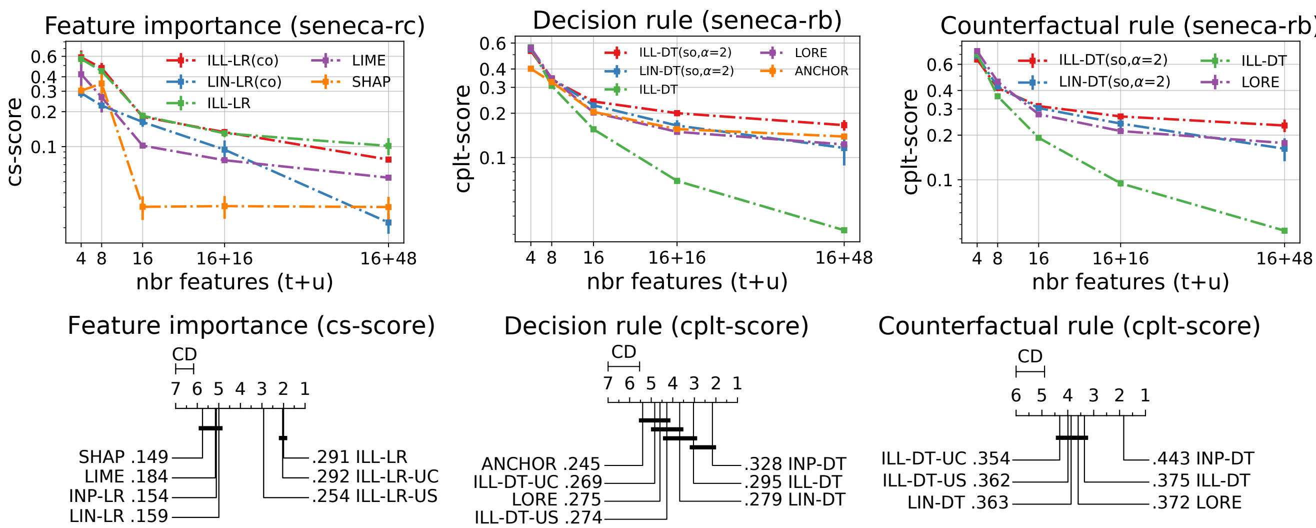

Fig. A4: Explanation correctness for synthetic classifiers. In the Figures, average metrics for correctness in feature importance and decision rules are reported varying the input feature size. In the CD plots, average rankings for individual metrics are reported for different methods and datasets.

TABLE A1: Feature-based latent space quality metrics for all datasets and methods.

	Feature Preservation						**KNN Gain**					
	ILL-UC	LIN-UC	PCA	ISOMAP	LLE	UMAP	ILL-UC	LIN-UC	PCA	ISOMAP	LLE	UMAP
aids	.842	.851	**1.000**	.706	.527	.683	.946	.960	**1.008**	.896	.791	.895
austr	.852	.855	**.993**	.790	.612	.732	**1.009**	1.001	1.000	.984	.960	.984
bank	.845	.843	**.952**	.786	.594	.666	.963	.963	.959	.938	.990	**.998**
breast	**1.000**	.996	**1.000**	.928	.746	.849	1.000	**1.010**	1.000	1.001	1.001	1.001
churn	.908	.861	**.919**	.668	.521	.679	1.078	1.026	**1.234**	.977	.960	1.000
compas	.859	.870	**1.000**	.724	.563	.643	.985	.994	1.002	1.019	.976	**1.020**
ctg	.939	.927	**1.000**	.716	.635	.678	**1.033**	1.023	1.000	1.000	.996	.996
diabetes	**1.000**	**1.000**	**1.000**	.845	.700	.792	**1.019**	1.000	1.000	.964	.848	.894
ecoli	**1.000**	**1.000**	**1.000**	.871	.806	.785	**1.024**	1.007	1.000	.977	.950	.961
fico	**1.000**	**1.000**	**1.000**	.832	.657	.680	**1.006**	1.003	1.000	.984	.964	.990
german	.849	.832	**.945**	.765	.589	.686	.994	.963	1.054	**1.067**	1.034	.992
home	**1.000**	**1.000**	**1.000**	.897	.679	.826	**1.033**	1.000	1.011	1.011	.956	.966
ionos	.994	.992	**1.000**	.868	.713	.749	**1.053**	1.045	1.052	1.006	1.004	1.006
sonar	.986	.967	**.995**	.871	.671	.800	1.003	.977	**1.027**	.890	.998	.946
spam	.921	.930	**.953**	.816	.585	.673	**1.016**	1.015	.994	.975	.924	.977
titanic	.868	.869	**1.000**	.896	.603	.797	**1.047**	**1.047**	1.000	1.014	.977	1.002
wine	**1.000**	**1.000**	**1.000**	.868	.670	.789	**1.063**	1.000	1.000	.974	.890	.933
yeast	.999	**1.000**	**1.000**	.887	.652	.799	.995	**1.000**	**1.000**	.965	.746	.870

LR and DT trained on latent spaces both with and without black-box decision conditioning. The same results are aggregated and ranked in Figure A5. Firstly, we observe that models without conditioning consistently achieve lower accuracies in surrogate classification tasks. Secondly, particularly for the LR surrogate, ILLUME demonstrates significant improvements compared to LIN encodings. This underscores the importance of label conditioning, combined with using more expressive latent features, in enhancing the accuracy of surrogate predictions.

RQ2 - Correctness for Synthetic Black-Box Explanations

In Figure A4 we report additional results from synthetic black-box models. We also present preliminary results on generating **counterfactual explanations** using ILLUME.

In the top pictures, we display with line plots the variation of explanation accuracy metrics with respect to the number of input dimensions, comparing the best-regularized setup of ILLUME with the un-regularized one. For feature importance explanations, we observe that the non-collinear (*co*) regularizer slightly enhances the correctness of the explanations across most scenarios. However, in cases involving 16+48 input features, the explanation accuracy experiences a minor decline. In contrast, for both factual and counterfactual rules, the combination of sparsity (α=2) and soft-orthogonality (*so*) significantly improves the correctness of the explanations. These findings underscore the importance of consistently applying sparsity regularizers within ILLUME when generating rule-based explanations.

In the CD plots, corresponding to the results presented for **RQ2** in the main paper, we display aggregated rankings for each metric across all datasets. We observe the same trends as reported in the main paper: ILLUME and its variants perform similarly in terms of feature importance correctness, while

ILLUME-DT shows comparable performance to INP-DT in factual rules correctness. Regarding counterfactual rules, there is no statistically significant difference between ILLUME-DT and LORE. However, INP-DT outperforms the others, consistent with its superior performance on factual rules.

RQ3 - Faithfulness and Robustness in Real-World Datasets Tables A6 and A7 present comprehensive results for feature importance metrics, while Tables A8 and A9 display detailed results for decision rules metrics. These tables include all datasets and black-box models that were summarized for answering **RQ3** in the main paper.

Figure A6 presents additional findings that compare the effects of different regularizations on the robustness and faithfulness metrics for feature importance explanations. We observe that sparse regularizations (so, α=2) do not enhance either robustness or faithfulness for feature importance. Additionally, models regularized for non-collinearity (co) perform similarly to unregularized models, although they achieve the highest average robustness scores. In contrast, Figures A7 show additional results comparing the impact of various regularizations on the robustness and faithfulness metrics for decision rules. Here, sparse regularizations (so, α=2) significantly improve the quality of decision rules for both metrics, thereby confirming our earlier observations regarding sparsity and decision rule correctness. Overall, for both types of explanations, models optimized for stability and label conditioning perform significantly worse in terms of robustness and faithfulness, respectively.

Tables A10 and Tables A11 present additional results on explainers' **robustness** with the evaluation of a **global metric** rather than local metrics based on sensitivity to perturbations. Intuitively, a –globally–robust explainer should produce similar explanations for closely located data points – with the same back-box predicted label – and distinct explanations for those farther apart. Thus, robustness is assessed globally by calculating the Spearman's rank correlation coefficient between the pairwise distances of explanations, $\{d_{\mathcal{E}}(i,j)\}_{i<j}$, and the corresponding pairwise distances of input records, $\{||\mathbf{x}_i - \mathbf{x}_j||_2)\}_{i<j}$, for those pairs of points with accordant predicted labels. Like in SENECA, we use similarity metrics $cs\text{-}score(\cdot,\cdot)$ and $cplt\text{-}score(\cdot,\cdot)$ instead of distance metrics. Figure A8 also presents the aggregate performance of feature importance explainers. In this analysis, ILLUME-LR achieves the highest performance, while ILLUME-LR-US performs comparably. This is in contrast to the local robustness results reported in the main paper, where ILLUME-LR-US was significantly worse. Additionally, LIME demonstrates the poorest performance in this context, despite showing adequate local robustness in the main paper. Regarding decision rule explainers, LIN-LR experiences a decline in performance compared to the local robustness results and ranks similarly to both ILLUME-DT and ILLUME-DT-US.

H. Additional Experimental Results on Synthetic Data

In this section, we further extend the evaluation of explanation methods on synthetic benchmarks beyond SENECA framework. In line with [12], we adopt synthetic nonlinear functions derived from the Friedman model [13] and adapted for classification tasks. Concretely, we employ publicly available datasets[14] `ssin-2c`, `int2-3c` and `int2-8c` respectively

TABLE A2: Ranking correctness of feature importance. Prediction accuracy of surrogate models inside parentheses.

	Feature Ranking Correctness (*spearman*)		
LGB	ssin-2c	int2-3c	int2-8p
ILL-LR(co)	**1.000**±.000 (98.6)	.184±.038 (92.3)	.204±.039 (81.7)
ILL-LR-UC(co)	**1.000**±.000 (98.5)	.185±.054 (91.8)	.202±.043 (81.7)
ILL-LR-US(co)	.904±.041 (98.7)	**.473**±.107 (96.3)	**.235**±.079 (82.1)
LIN-LR(co)	.836±.009 (97.2)	.033±.001 (89.1)	.117±.013 (80.4)
INP-LR	.806±.006 (96.6)	.032±.001 (89.0)	.110±.012 (80.1)
LIME	.993±.001	.009±.001	.082±.003
SHAP	.257±.001	.197±.010	.163±.020
XGB	ssin-2c	int2-3c	int2-8p
ILL-LR(co)	**1.000**±.000 (98.0)	.264±.043 (91.6)	.197±.019 (82.7)
ILL-LR-UC(co)	**1.000**±.000 (98.3)	.186±.055 (91.8)	.199±.039 (82.7)
ILL-LR-US(co)	.947±.028 (98.1)	**.676**±.095 (96.0)	**.458**±.109 (83.3)
LIN-LR(co)	.847±.008 (97.0)	.036±.004 (89.0)	.117±.012 (81.4)
INP-LR	.812±.009 (96.1)	.033±.001 (88.7)	.108±.013 (81.2)
LIME	.995±.002	.010±.001	.081±.005
SHAP	.276±.008	.211±.007	.155±.018
CTB	ssin-2c	int2-3c	int2-8p
ILL-LR(co)	**1.000**±.000 (98.6)	.165±.048 (90.6)	**.278**±.021 (82.7)
ILL-LR-UC(co)	**1.000**±.000 (98.4)	.168±.055 (91.1)	.203±.042 (82.7)
ILL-LR-US(co)	.965±.015 (98.6)	**.542**±.050 (97.2)	.147±.025 (82.9)
LIN-LR(co)	.824±.006 (97.3)	.038±.006 (89.0)	.116±.011 (81.6)
INP-LR	.806±.006 (96.9)	.033±.002 (88.9)	.111±.011 (81.4)
LIME	.993±.001	.008±.001	.084±.002
SHAP	.259±.012	.218±.009	.122±.007

TABLE A3: Correctness of synthetic explanations. Prediction accuracy of surrogate models inside parentheses.

	Feature Importance Correctness (*cs-score*)		
LGB	ssin-2c	int2-3c	int2-8p
ILL-LR(co)	**.998**±.000 (98.6)	.139±.042 (92.3)	.210±.012 (81.7)
ILL-LR-UC(co)	**.998**±.000 (98.5)	.123±.050 (91.8)	.211±.017 (81.7)
ILL-LR-US(co)	.963±.011 (98.7)	**.332**±.027 (96.3)	**.305**±.074 (82.1)
LIN-LR(co)	.899±.004 (97.2)	.050±.003 (89.1)	.249±.012 (80.4)
INP-LR	.891±.004 (96.6)	.047±.003 (89.0)	.248±.012 (80.1)
LIME	.993±.001	.000±.000	.259±.017
SHAP	.223±.011	.315±.015	.142±.021
XGB	ssin-2c	int2-3c	int2-8p
ILL-LR(co)	**.996**±.001 (98.0)	.176±.050 (91.6)	.219±.021 (82.7)
ILL-LR-UC(co)	**.997**±.001 (98.3)	.124±.054 (91.8)	.213±.020 (82.7)
ILL-LR-US(co)	.967±.010 (98.1)	**.411**±.028 (96.0)	**.305**±.104 (83.3)
LIN-LR(co)	.903±.005 (97.0)	.053±.005 (89.0)	.254±.014 (81.4)
INP-LR	.893±.005 (96.1)	.049±.002 (88.7)	.254±.013 (81.2)
LIME	.992±.001	.000±.000	.269±.017
SHAP	.236±.006	.321±.008	.137±.023
CTB	ssin-2c	int2-3c	int2-8p
ILL-LR(co)	**.998**±.001 (98.6)	.100±.050 (90.6)	.221±.012 (82.7)
ILL-LR-UC(co)	**.998**±.001 (98.4)	.117±.052 (91.1)	.211±.017 (82.7)
ILL-LR-US(co)	.989±.003 (98.6)	**.357**±.015 (97.2)	.230±.058 (82.9)
LIN-LR(co)	.899±.005 (97.3)	.054±.007 (89.0)	**.251**±.013 (81.6)
INP-LR	.891±.004 (96.9)	.044±.001 (88.9)	**.250**±.013 (81.4)
LIME	.993±.001	.000±.000	.244±.012
SHAP	.231±.014	.308±.004	.082±.011

[14]http://www3.dsi.uminho.pt/pcortez/data

with two, three and eight classes. They consists of $n = 1000$ synthetic instances described by 4 continuous features $\{x_1, x_2, x_3, x_4\}$. Instead of providing ground-truth importance values $\{\hat{\psi}_1, \hat{\psi}_2, \hat{\psi}_3, \hat{\psi}_4\}$, in [12] the authors provides importance rankings: in `ssin-2c`, the ranking is $\hat{\psi}_1 > \hat{\psi}_2 > \hat{\psi}_3 > \hat{\psi}_4$; in `int2-3c` and `int2-8c`, the ranking is $(\hat{\psi}_1, \hat{\psi}_2) > \hat{\psi}_3 > \hat{\psi}_4$. Using the same experimental setting adopted for the SENECA benchmarks, we evaluate local explanation methods with respect to ranking quality (in Table A2 via Spearman correlation) and importance proximity (in Table A3 via the already used *cs-score*) across all test instances and five independent train–test splits in every dataset. To evaluate correctness scores, we set ground-truth importance values aligned with reference rankings as following: in `ssin-2c`, we define $\hat{\psi} = \{4., 2., 1., 0.\}$; in `int2-3c` and `int2-8c`, we define $\hat{\psi} = \{3., 3., 1., 0.\}$. Because these synthetic datasets do not provide white-box predictive functions like in SENECA, we fit ensemble-tree black-boxes to mimic the underlying decision process, and then explain their local predictions. In multi-class datasets, explanations are generated for the majority class. The reported results reveal that ILLUME consistently outperforms competing explainers. On `ssin-2c`, ILL-LR and ILL-LR-UC achieve perfect rankings and similarities, with LIME as the second best approach. For the `int2` datasets, ILL-LR and ILL-LR-US mostly attains the best results with XGB and LGB, consistently outperforming LIME and treeSHAP. When CTB is explained, however, the global linear baselines LIN-LR and INP-LR yield more aligned explanations on `int2-8p`. Overall, these findings underscore the value of carefully designing synthetic benchmarks for explanation methods evaluation.

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TABLE A4: Decision-based latent space quality metrics with LGB as black-box for all datasets and methods.

LGB	**Decision Preservation**				**LR Accuracy** (Macro-F1)				**DT Accuracy** (Macro-F1)			
	ILL	LIN	ILL-UC	LIN-UC	ILL	LIN	ILL-UC	LIN-UC	ILL	LIN	ILL-UC	LIN-UC
aids	**.618**	.596	.549	.555	**.952**	.902	.899	.902	**.930**	.874	.768	.780
austr	**.743**	.729	.613	.623	.920	**.927**	.919	.920	.927	**.942**	.869	.913
bank	**.659**	.656	.595	.588	**.901**	.866	.839	.840	**.881**	**.881**	.759	.736
breast	**.779**	.739	.728	.718	.981	.990	.981	**.991**	.972	.963	.981	**.991**
churn	**.610**	.583	.542	.548	**.875**	.637	.650	.623	**.852**	.827	.738	.760
compas	**.667**	.665	.644	.640	**.926**	.914	.902	.914	**.913**	.893	.891	.891
ctg	**.693**	.689	.624	.611	.993	**.996**	**.996**	.993	.993	**.996**	.969	.949
diabetes	**.642**	.640	.621	.627	**.906**	.871	.881	.871	**.874**	.832	.836	.864
ecoli	**.726**	.700	.706	.706	**.970**	.936	.947	.936	**.939**	**.939**	.910	.921
fico	**.642**	.578	.586	.580	**.908**	.903	.894	.903	.862	**.869**	.843	.854
german	**.669**	.627	.566	.557	.814	.807	**.816**	.806	**.805**	.791	.699	.707
home	**.677**	.626	.614	.614	**.949**	**.949**	**.949**	**.949**	**.970**	.939	.929	.949
ionos	.631	**.665**	.586	.597	.886	.906	**.934**	.854	.919	.952	**.968**	.907
sonar	**.671**	.629	.576	.610	**.904**	.857	.810	.833	**.881**	.833	.785	.786
spam	**.704**	.647	.674	.623	**.963**	.942	.942	.936	**.955**	.946	.913	.903
titanic	.743	**.744**	.650	.672	.885	**.893**	**.893**	.885	**.937**	.919	.922	.931
wine	**.550**	.546	.536	.540	**.305**	.266	.292	.286	.516	**.537**	.497	.440
yeast	**.630**	.615	.609	.623	**.629**	.565	.612	.572	**.556**	.534	.525	.512

TABLE A5: Decision-based latent space quality metrics with XGB as black-box for all datasets and methods.

XGB	**Decision Preservation**				**LR Accuracy** (Macro-F1)				**DT Accuracy** (Macro-F1)			
	ILL	LIN	ILL-UC	LIN-UC	ILL	LIN	ILL-UC	LIN-UC	ILL	LIN	ILL-UC	LIN-UC
aids	**.661**	.610	.563	.557	**.941**	.871	.880	.868	**.938**	.877	.730	.774
austr	.739	**.751**	.657	.652	**.934**	.913	.912	.912	**.941**	.911	.898	.898
bank	.653	**.662**	.593	.582	**.893**	.868	.832	.853	**.886**	.873	.750	.744
breast	**.821**	.737	.775	.730	.981	**.990**	.981	**.990**	.972	.972	.961	**1.000**
churn	**.602**	.587	.536	.541	**.872**	.660	.649	.644	**.852**	.786	.728	.728
compas	.671	**.673**	.640	.638	**.933**	.915	.911	.915	**.915**	.912	.891	.888
ctg	**.743**	.739	.653	.629	**1.000**	.996	.996	.996	.957	.945	**1.000**	.945
diabetes	**.684**	.649	.625	.626	**.910**	.871	.879	.871	.857	**.895**	.849	.857
ecoli	.721	.703	**.726**	.709	**1.000**	.969	.988	.980	.980	**.992**	.947	.980
fico	**.650**	.588	.592	.588	**.937**	.929	.922	.929	.886	**.897**	.882	.874
german	**.655**	.617	.535	.516	.789	**.810**	.761	.778	.780	**.791**	.714	.697
home	**.663**	.642	.630	.626	.939	**.949**	.939	.939	.929	**.970**	.939	.929
ionos	.566	**.603**	.575	.572	.891	.868	**.904**	.883	.874	**.954**	.936	.880
sonar	.619	**.629**	.595	**.629**	.786	.785	.810	**.833**	**.857**	.762	.786	.762
spam	**.721**	.667	.675	.631	**.968**	.947	.942	.940	**.966**	.952	.919	.901
titanic	**.725**	.687	.655	.694	.877	**.890**	.883	.877	**.923**	.921	.921	.910
wine	**.587**	.574	.562	.566	**.334**	.276	.319	.287	.422	**.544**	.441	.494
yeast	**.700**	.695	.690	.694	**.790**	.755	.776	.752	.676	**.698**	.644	.656

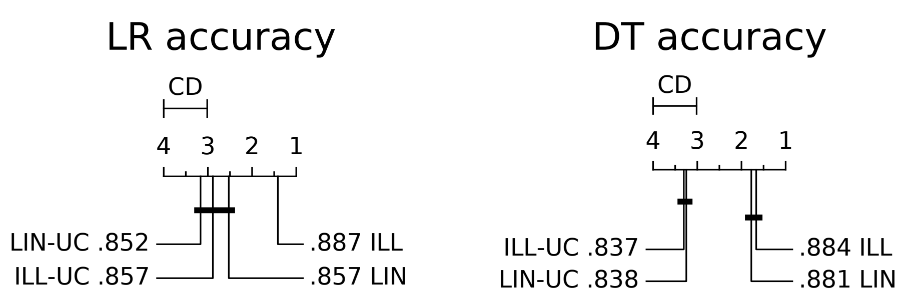

Fig. A5: Critical Difference plots with Nemenyi test at 90% confidence level for prediction accuracy metrics in real-world datasets, comparing LR and DT performances over different training feature spaces.

TABLE A6: Robustness and faithfulness metrics for feature importance methods in all datasets with LGB as black-box. Prediction accuracy of surrogate classifiers is reported inside parentheses.

LGB	**Feature Importance Robustness**						**Feature Importance Faithfulness**					
	ILL-LR	ILL-LR-US	LIN-LR	INP-LR	LIME	SHAP	ILL-LR	ILL-LR-UC	LIN-LR	INP-LR	LIME	SHAP
aids	**.994** (95.3)	.797 (97.0)	.582 (93.7)	.244 (93.7)	<u>.978</u>	.445	**.908** (97.4)	.234 (93.5)	.540 (93.5)	.610 (93.7)	.054	<u>.806</u>
austr	**.980** (93.5)	.901 (89.9)	.659 (92.0)	.500 (92.0)	<u>.904</u>	.443	<u>.785</u> (92.0)	.520 (92.0)	.561 (89.9)	.659 (92.0)	.033	**.798**
bank	**.994** (96.1)	<u>.958</u> (96.0)	.536 (94.9)	.271 (95.0)	.916	.127	**.790** (96.8)	.304 (95.4)	.537 (95.0)	<u>.540</u> (95.0)	.122	.410
breast	.777 (98.2)	<u>.811</u> (97.4)	.388 (99.1)	.219 (98.2)	**.964**	.328	.846 (99.1)	<u>.859</u> (98.2)	.819 (97.4)	.833 (98.2)	.113	**.906**
churn	**.981** (95.8)	<u>.932</u> (94.8)	.719 (87.3)	.672 (87.6)	.618	.160	**.679** (96.6)	.142 (89.1)	.055 (87.4)	.247 (87.6)	.091	<u>.644</u>
compas	**.993** (94.7)	.711 (95.1)	.711 (94.4)	.635 (94.4)	<u>.971</u>	.538	**.798** (95.8)	.317 (94.0)	.319 (94.4)	.313 (94.4)	.015	<u>.448</u>
ctg	**.980** (99.8)	<u>.840</u> (99.8)	.758 (99.5)	.617 (99.3)	.781	.656	**.741** (99.8)	.357 (99.5)	.624 (99.5)	.519 (99.3)	.487	<u>.682</u>
diabetes	<u>.966</u> (92.9)	.797 (94.2)	.180 (89.6)	.176 (89.6)	**.985**	.163	.396 (92.9)	.298 (90.9)	<u>.454</u> (89.6)	.450 (89.6)	.021	**.580**
ecoli	.958 (100.)	<u>.961</u> (100.)	.735 (100.)	.673 (100.)	**.981**	.585	**.640** (100.)	.623 (100.)	.575 (100.)	.586 (100.)	.260	<u>.503</u>
fico	**.981** (90.3)	.899 (90.8)	.400 (90.3)	.324 (90.3)	<u>.918</u>	.268	<u>.443</u> (91.4)	.338 (89.5)	.379 (90.3)	.377 (90.3)	.000	**.600**
german	**.996** (89.0)	<u>.930</u> (85.5)	.394 (85.5)	.338 (88.5)	.840	.069	**.531** (87.5)	.198 (88.0)	.174 (85.5)	.190 (88.5)	.124	<u>.238</u>
home	<u>.878</u> (96.0)	.756 (97.0)	.204 (94.9)	.122 (94.9)	**.976**	.178	<u>.741</u> (97.0)	.337 (94.9)	.495 (94.9)	.486 (94.9)	.077	**.752**
ionos	<u>.845</u> (94.4)	.688 (94.4)	.371 (88.7)	.353 (88.7)	**.932**	.366	.385 (93.0)	.153 (94.4)	.568 (91.5)	<u>.576</u> (88.7)	.193	**.841**
sonar	.621 (85.7)	<u>.698</u> (92.9)	.104 (83.3)	.085 (83.3)	**.914**	.078	<u>.211</u> (90.5)	.055 (88.1)	.201 (88.1)	.077 (83.3)	.032	**.469**
spam	**.923** (96.0)	<u>.859</u> (96.2)	.396 (94.0)	.311 (94.4)	.808	.166	**.848** (96.4)	.441 (94.5)	.522 (94.6)	.037 (94.4)	.000	<u>.662</u>
titanic	**.991** (90.5)	<u>.919</u> (90.5)	.616 (89.9)	.646 (89.4)	.971	.536	**.769** (91.6)	.640 (90.5)	.639 (89.9)	.641 (89.4)	.389	<u>.733</u>
wine	**.943** (64.2)	<u>.640</u> (67.2)	.192 (59.1)	.176 (58.6)	.779	.166	**.152** (72.3)	.105 (62.8)	.135 (58.8)	<u>.146</u> (58.6)	.022	.104
yeast	**.934** (76.4)	<u>.705</u> (75.1)	.323 (69.4)	.305 (69.4)	.289	.128	**.280** (78.5)	.143 (74.1)	.216 (70.7)	<u>.217</u> (69.4)	.018	.177

TABLE A7: Robustness and faithfulness metrics for feature importance methods in all datasets with XGB as black-box. Prediction accuracy of surrogate classifiers is reported inside parentheses.

XGB	**Feature Importance Robustness**						**Feature Importance Faithfulness**					
	ILL-LR	ILL-LR-US	LIN-LR	INP-LR	LIME	SHAP	ILL-LR	ILL-LR-UC	LIN-LR	INP-LR	LIME	SHAP
aids	**.994** (95.3)	.901 (95.1)	.505 (91.1)	.328 (91.1)	<u>.965</u>	.393	**.838** (95.8)	.252 (91.8)	.654 (91.1)	.574 (91.1)	.083	<u>.799</u>
austr	**.987** (92.0)	.872 (92.8)	.614 (90.6)	.514 (90.6)	<u>.929</u>	.341	**.783** (94.2)	.560 (91.3)	.704 (90.6)	.698 (90.6)	.000	<u>.737</u>
bank	**.989** (96.2)	<u>.929</u> (96.0)	.503 (95.3)	.260 (95.5)	.910	.213	**.826** (97.1)	.397 (94.9)	.352 (93.8)	.473 (95.5)	.269	<u>.765</u>
breast	<u>.875</u> (99.1)	.803 (99.1)	.451 (100.)	.502 (97.4)	**.970**	.302	<u>.888</u> (98.2)	.883 (98.2)	.850 (98.2)	.864 (97.4)	.128	**.927**
churn	**.974** (95.1)	<u>.840</u> (95.1)	.776 (88.0)	.680 (87.9)	.549	.163	**.612** (95.8)	.128 (89.1)	.154 (88.5)	.197 (87.9)	.078	<u>.611</u>
compas	**.989** (95.4)	.742 (95.2)	.713 (94.5)	.597 (94.5)	<u>.971</u>	.532	**.731** (96.1)	.320 (94.4)	.330 (94.5)	.321 (94.5)	.004	<u>.525</u>
ctg	**.992** (100.)	<u>.859</u> (99.8)	.670 (99.8)	.477 (99.8)	.824	.502	**.776** (100.)	.323 (99.8)	.688 (99.8)	.434 (99.8)	.066	<u>.696</u>
diabetes	<u>.971</u> (92.2)	.780 (92.9)	.225 (90.3)	.208 (87.0)	**.984**	.171	.390 (92.9)	.273 (90.3)	<u>.517</u> (89.0)	.507 (87.0)	.048	**.671**
ecoli	<u>.973</u> (100.)	.945 (100.)	.722 (100.)	.659 (98.5)	**.990**	.726	**.720** (100.)	.711 (100.)	.683 (100.)	.691 (98.5)	.358	<u>.516</u>
fico	**.973** (93.4)	.944 (93.2)	.433 (92.9)	.368 (92.9)	<u>.971</u>	.304	<u>.516</u> (95.3)	.409 (92.3)	.453 (92.9)	.450 (92.9)	.000	**.672**
german	**.996** (87.0)	<u>.953</u> (87.0)	.512 (86.0)	.320 (86.0)	.782	.071	**.492** (89.0)	.131 (86.5)	.167 (82.5)	.200 (86.0)	.005	<u>.358</u>
home	<u>.956</u> (93.9)	.912 (96.0)	.162 (93.9)	.141 (93.9)	**.974**	.148	<u>.636</u> (97.0)	.277 (94.9)	.474 (93.9)	.470 (93.9)	.027	**.712**
ionos	<u>.772</u> (91.5)	.734 (91.5)	.382 (87.3)	.351 (90.1)	**.950**	.322	.244 (97.2)	.189 (91.5)	.533 (88.7)	<u>.551</u> (90.1)	.162	**.914**
sonar	<u>.784</u> (81.0)	.686 (81.0)	.090 (76.2)	.069 (78.6)	**.935**	.168	<u>.183</u> (83.3)	.137 (83.3)	.124 (81.0)	.084 (78.6)	.042	**.633**
spam	**.912** (96.5)	.819 (96.9)	.291 (94.5)	.141 (94.6)	<u>.878</u>	.140	**.823** (97.0)	.462 (94.5)	.421 (94.6)	.265 (94.6)	.013	<u>.721</u>
titanic	**.993** (91.1)	.921 (91.1)	.662 (89.4)	.608 (89.9)	<u>.973</u>	.633	.571 (90.5)	<u>.671</u> (89.9)	.638 (88.8)	.634 (89.9)	.155	**.809**
wine	**.952** (66.3)	.581 (66.6)	.183 (60.4)	.174 (59.3)	<u>.856</u>	.278	**.223** (72.9)	.165 (64.3)	.206 (60.1)	<u>.209</u> (59.3)	.042	.107
yeast	**.964** (86.5)	.795 (88.2)	.391 (81.5)	.394 (81.8)	<u>.941</u>	.813	**.528** (88.2)	.353 (84.2)	.325 (81.5)	.311 (81.8)	.152	<u>.290</u>

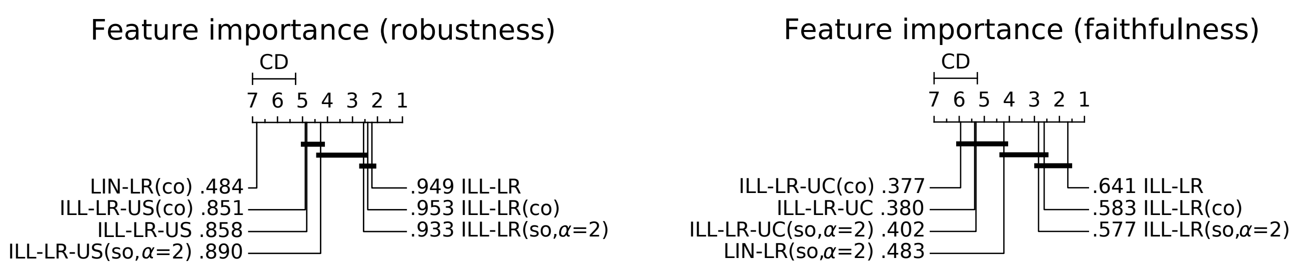

Fig. A6: Critical Difference plots with Nemenyi test at 90% confidence level for robustness and faithfulness metrics in real-world datasets, comparing various regularized versions of ILLUME-LR.

TABLE A8: Robustness and faithfulness metrics for decision rules methods in all datasets with LGB as black-box. Prediction accuracy of surrogate classifiers is reported inside parentheses.

LGB	**Decision Rule Robustness**						**Decision Rule Faithfulness**					
	ILL-DT	ILL-DT-US	LIN-DT	INP-DT	LORE	ANCHOR	ILL-DT	ILL-DT-UC	LIN-DT	INP-DT	LORE	ANCHOR
aids	.676 (97.4)	.301 (96.7)	**.911** (97.9)	.486 (98.8)	.060	.180	.739 (97.4)	.644 (96.5)	.825 (97.9)	.804 (98.8)	.762	**.868**
austr	.404 (92.0)	.438 (92.0)	**.573** (97.1)	.288 (94.9)	.147	.218	**.707** (92.0)	.653 (94.9)	.694 (97.1)	.661 (94.9)	.666	.672
bank	.595 (95.5)	.504 (95.5)	**.679** (95.3)	.600 (96.1)	.038	.053	.642 (95.5)	.566 (93.8)	**.734** (95.3)	.675 (96.1)	.510	.457
breast	.352 (98.2)	.114 (96.5)	.452 (98.2)	**.536** (96.5)	.077	.025	**.863** (98.2)	.788 (98.2)	.817 (96.5)	.736 (96.5)	.735	.381
churn	.618 (95.7)	.471 (95.2)	**.890** (97.3)	.358 (98.2)	.215	.099	**.609** (95.7)	.399 (91.6)	.560 (97.3)	.605 (98.2)	.605	.413
compas	.603 (95.1)	.351 (94.3)	**.961** (95.1)	.361 (95.8)	.115	.168	.584 (95.1)	.564 (94.5)	**.619** (95.1)	.536 (95.8)	.503	.592
ctg	.791 (99.3)	.677 (99.3)	**.975** (99.3)	.866 (99.3)	.317	.031	.777 (99.3)	.779 (99.1)	.743 (99.3)	**.839** (99.3)	.666	.335
diabetes	.116 (89.6)	.057 (87.7)	**.166** (89.6)	.098 (88.3)	.047	.097	.449 (89.6)	.545 (90.9)	.344 (89.6)	.295 (88.3)	.276	**.646**
ecoli	.632 (100.)	.569 (100.)	**.784** (100.)	.529 (100.)	.478	.462	.570 (100.)	.587 (100.)	.531 (100.)	.513 (100.)	.541	**.600**
fico	.374 (90.1)	.262 (89.5)	**.390** (89.5)	.200 (90.5)	.038	.167	.512 (90.1)	.425 (89.4)	.512 (89.5)	.481 (90.5)	.457	**.637**
german	**.618** (86.0)	.287 (85.5)	.562 (83.0)	.154 (81.0)	.058	.085	.327 (86.0)	.142 (87.0)	.112 (83.0)	.141 (81.0)	.118	**.369**
home	.064 (96.0)	.058 (96.0)	.051 (98.0)	**.150** (94.9)	.014	.073	.511 (96.0)	.420 (97.0)	.469 (98.0)	.485 (94.9)	.407	**.680**
ionos	.438 (98.6)	.297 (97.2)	**.570** (95.8)	.376 (95.8)	.204	.155	**.793** (98.6)	.722 (97.2)	.751 (95.8)	.697 (95.8)	.748	.645
sonar	.194 (83.3)	.147 (85.7)	**.515** (81.0)	.398 (83.3)	.108	.043	**.458** (85.7)	.444 (85.7)	.422 (81.0)	.453 (83.3)	.214	.441
spam	.402 (96.1)	.459 (95.1)	**.652** (94.5)	.249 (93.7)	.120	.113	**.767** (95.1)	.622 (92.8)	.701 (94.5)	.642 (93.7)	.253	.495
titanic	.628 (93.9)	.476 (95.0)	**.948** (96.1)	.687 (94.4)	.288	.420	.762 (95.0)	.748 (96.1)	.792 (96.1)	.702 (94.4)	.697	**.795**
wine	.296 (72.2)	.230 (70.1)	**.525** (70.6)	.154 (70.2)	.134	.224	.097 (72.2)	.087 (71.4)	.099 (70.6)	.108 (70.2)	.214	**.388**
yeast	.568 (78.5)	.471 (77.8)	**.713** (76.4)	.362 (78.5)	.467	.418	.412 (78.5)	.367 (77.8)	.274 (75.4)	.319 (78.5)	**.427**	.358

TABLE A9: Robustness and faithfulness metrics for decision rules methods in all datasets with XGB as black-box. Prediction accuracy of surrogate classifiers is reported inside parentheses.

XGB	**Decision Rule Robustness**						**Decision Rule Faithfulness**					
	ILL-DT	ILL-DT-US	LIN-DT	INP-DT	LORE	ANCHOR	ILL-DT	ILL-DT-UC	LIN-DT	INP-DT	LORE	ANCHOR
aids	.666 (95.6)	.378 (93.0)	**.798** (94.2)	.098 (95.1)	.023	.153	.642 (95.6)	.518 (93.7)	.657 (94.2)	.645 (95.1)	.639	**.821**
austr	.513 (92.0)	.387 (92.8)	**.713** (92.8)	.262 (89.9)	.123	.256	**.780** (92.8)	.691 (90.6)	.587 (92.8)	.591 (89.9)	.697	.751
bank	.620 (95.9)	.528 (95.5)	**.767** (94.9)	.549 (95.1)	.022	.047	.728 (95.9)	.539 (95.6)	.551 (94.9)	**.743** (95.1)	.360	.465
breast	.243 (96.5)	.184 (96.5)	.465 (97.4)	**.563** (96.5)	.068	.032	**.813** (96.5)	.772 (97.4)	.769 (97.4)	.795 (96.5)	.703	.374
churn	.291 (96.1)	.625 (94.5)	**.895** (95.8)	.355 (99.0)	.229	.112	.517 (96.1)	.365 (93.4)	.504 (95.8)	**.571** (99.0)	.533	.394
compas	.811 (94.0)	.448 (94.2)	**.965** (94.5)	.385 (94.6)	.101	.154	**.606** (94.2)	.523 (93.4)	.576 (94.5)	.506 (94.6)	.489	.592
ctg	.869 (99.8)	.852 (100.)	**.976** (99.5)	.882 (99.5)	.340	.033	.787 (100.)	.727 (99.3)	.722 (99.5)	**.795** (99.5)	.654	.347
diabetes	.124 (91.6)	.062 (90.3)	.124 (90.9)	.176 (92.2)	.044	**.177**	.532 (91.6)	.563 (92.2)	.503 (90.9)	.411 (92.2)	.293	**.688**
ecoli	.680 (100.)	.533 (100.)	**.800** (100.)	.588 (100.)	.581	.497	.748 (100.)	.673 (100.)	**.753** (100.)	.666 (100.)	.623	.562
fico	.340 (93.8)	.279 (92.1)	**.413** (92.8)	.221 (93.8)	.053	.187	.617 (93.8)	.632 (93.0)	.614 (92.8)	.628 (93.8)	.643	**.679**
german	.337 (87.0)	.129 (82.5)	**.565** (81.5)	.107 (83.5)	.033	.081	.226 (87.0)	.169 (81.0)	.052 (81.5)	.041 (83.5)	.098	**.388**
home	.095 (96.0)	.037 (96.0)	.082 (97.0)	**.132** (94.9)	.010	.062	.578 (96.0)	.414 (97.0)	.555 (97.0)	.488 (94.9)	.455	**.730**
ionos	.385 (95.8)	.374 (94.4)	.514 (97.2)	**.615** (97.2)	.212	.232	.701 (95.8)	.683 (95.8)	.790 (97.2)	**.829** (97.2)	.710	.730
sonar	.211 (85.7)	.221 (85.7)	**.533** (88.1)	.254 (92.9)	.141	.072	.546 (85.7)	**.597** (85.7)	.324 (88.1)	.535 (92.9)	.427	.516
spam	.419 (96.1)	.483 (97.1)	**.699** (94.6)	.403 (94.7)	.130	.090	**.797** (97.1)	.643 (93.1)	.663 (94.6)	.675 (94.7)	.227	.478
titanic	.620 (95.5)	.469 (95.0)	**.955** (97.2)	.693 (96.1)	.387	.463	**.819** (95.5)	.690 (97.2)	.781 (97.2)	.737 (96.1)	.738	.807
wine	.303 (71.0)	.244 (71.0)	**.458** (70.6)	.162 (68.2)	.150	.229	.160 (71.0)	.148 (71.5)	.164 (70.6)	.145 (68.2)	.298	**.403**
yeast	.533 (89.2)	.437 (89.9)	**.788** (88.2)	.427 (89.9)	.469	.387	.590 (89.9)	.469 (87.5)	.470 (88.2)	.538 (89.9)	**.613**	.361

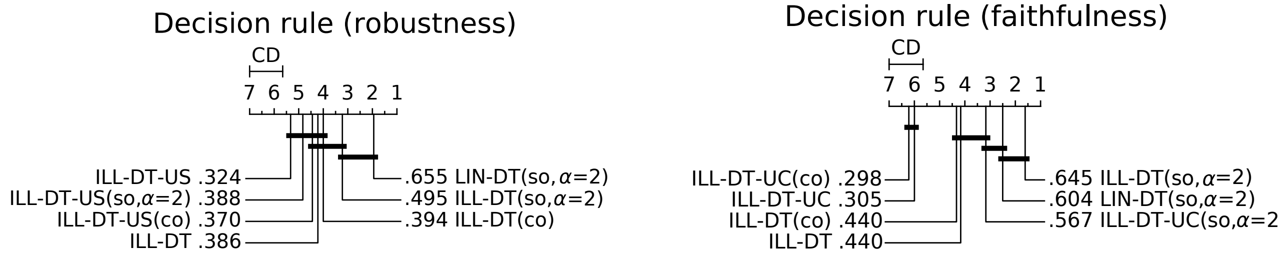

Fig. A7: Critical Difference plots with Nemenyi test at 90% confidence level for robustness and faithfulness metrics in real-world datasets, comparing various regularized versions of ILLUME-DT.

TABLE A10: Global robustness metrics in all datasets with LGB as black-box. Prediction accuracy of surrogate classifiers is reported inside parentheses.

LGB	Feature Importance Robustness						Decision Rule Robustness					
	ILL-LR	ILL-LR-US	LIN-LR	INP-LR	LIME	SHAP	ILL-DT	ILL-DT-UC	LIN-DT	INP-DT	LORE	ANCHOR
aids	**.743** (95.3)	.653 (97.0)	.563 (93.7)	.352 (93.7)	.163	.254	.267 (97.4)	.252 (96.7)	**.595** (97.9)	.192 (98.8)	.170	.191
australian	**.801** (93.5)	.651 (89.9)	.488 (92.0)	.396 (92.0)	.041	.181	.297 (91.3)	**.383** (92.0)	.361 (97.1)	.274 (94.9)	.197	.141
bank	**.897** (96.1)	.839 (96.0)	.798 (94.9)	.671 (95.0)	.147	.298	**.540** (95.5)	.387 (95.5)	.425 (95.3)	.192 (96.1)	.055	.066
breast	.532 (97.4)	**.638** (96.5)	.568 (99.1)	.553 (98.2)	.280	.226	**.285** (98.2)	.266 (96.5)	.227 (96.5)	.123 (96.5)	.024	.122
churn	**.664** (95.8)	.655 (94.8)	.614 (87.0)	.476 (87.6)	.000	.299	**.350** (95.7)	.256 (95.2)	.210 (97.3)	.167 (98.2)	.138	.100
compas	**.875** (94.7)	.685 (95.1)	.527 (94.4)	.421 (94.4)	.142	.352	.432 (95.1)	**.547** (94.3)	.493 (95.1)	.205 (95.8)	.193	.152
ctg	**.758** (99.8)	.729 (99.8)	.732 (99.5)	.647 (99.3)	.117	.405	.293 (99.3)	.445 (99.3)	**.511** (99.3)	.062 (99.3)	.011	.275
diabetes	**.700** (92.9)	.684 (94.2)	.499 (89.6)	.498 (89.6)	.486	.335	.305 (89.6)	.304 (87.7)	.255 (89.6)	**.344** (88.3)	.198	.231
ecoli	.825 (100.)	**.871** (100.)	.834 (100.)	.838 (100.)	.447	.749	.524 (100.)	**.653** (100.)	.451 (100.)	.277 (100.)	.456	.633
fico	**.744** (90.3)	.678 (90.8)	.501 (90.3)	.426 (90.3)	.415	.411	**.318** (90.1)	.300 (89.5)	.274 (89.5)	.224 (90.5)	.131	.247
german	**.736** (89.0)	.653 (85.5)	.559 (85.5)	.523 (88.5)	.072	.242	.243 (86.0)	.252 (85.5)	**.354** (83.0)	.214 (81.0)	.193	.251
home	**.720** (96.0)	.612 (97.0)	.606 (94.9)	.590 (94.9)	.437	.311	.478 (96.0)	.341 (96.0)	.430 (98.0)	**.510** (94.9)	.434	.449
ionosphere	**.741** (94.4)	.740 (94.4)	.723 (88.7)	.519 (88.7)	.259	.586	.630 (98.6)	**.663** (90.1)	.581 (95.8)	.144 (95.8)	.054	.212
sonar	**.793** (85.7)	.735 (92.9)	.725 (83.3)	.715 (83.3)	.288	.388	**.581** (83.3)	.382 (85.7)	.379 (81.0)	.228 (83.3)	.129	.102
spam	.386 (96.0)	.178 (96.2)	**.416** (94.0)	.357 (94.4)	.274	.069	.201 (94.7)	.273 (95.1)	.152 (94.5)	.097 (93.7)	.147	**.289**
titanic	**.864** (90.5)	.770 (90.5)	.777 (89.9)	.689 (89.4)	.551	.681	.612 (93.9)	.616 (95.0)	**.658** (96.1)	.569 (94.4)	.470	.548
wine	**.714** (64.2)	.602 (67.2)	.577 (59.1)	.554 (58.6)	.243	.484	.269 (72.2)	.271 (70.1)	.225 (70.6)	.253 (70.2)	.270	**.359**
yeast	**.617** (76.4)	.474 (75.1)	.436 (69.4)	.440 (69.4)	.096	.299	.331 (78.5)	.355 (77.8)	.380 (75.4)	.376 (78.5)	.248	**.411**

TABLE A11: Global robustness metrics in all datasets with XGB as black-box. Prediction accuracy of surrogate classifiers is reported inside parentheses.

XGB	Feature Importance Robustness						Decision Rule Robustness					
	ILL-LR	ILL-LR-US	LIN-LR	INP-LR	LIME	SHAP	ILL-DT	ILL-DT-UC	LIN-DT	INP-DT	LORE	ANCHOR
aids	**.720** (95.3)	.492 (95.1)	.485 (91.1)	.400 (91.1)	.159	.250	.298 (95.6)	**.400** (92.3)	.249 (94.2)	.196 (95.1)	.194	.186
australian	**.781** (92.0)	.482 (92.8)	.554 (90.6)	.569 (90.6)	.022	.269	**.378** (92.0)	.377 (92.8)	.291 (92.8)	.193 (89.9)	.262	.262
bank	**.872** (96.2)	.810 (96.0)	.758 (95.3)	.668 (95.5)	.105	.293	**.530** (95.9)	.496 (95.5)	.436 (94.9)	.145 (95.1)	.012	.055
breast	.698 (99.1)	**.707** (99.1)	.506 (100.0)	.272 (97.4)	.262	.251	.305 (96.5)	**.356** (95.6)	.295 (97.4)	.183 (96.5)	.127	.108
churn	.585 (95.1)	**.648** (95.1)	.604 (87.6)	.473 (87.9)	.017	.339	.324 (96.1)	**.326** (94.5)	.315 (95.8)	.192 (99.0)	.132	.175
compas	**.844** (95.4)	.713 (95.2)	.589 (94.5)	.407 (94.5)	.127	.354	.327 (94.0)	.546 (94.2)	**.552** (94.5)	.179 (94.6)	.162	.137
ctg	**.811** (100.)	.776 (99.8)	.760 (99.8)	.552 (99.8)	.000	.545	.402 (99.8)	.390 (100.)	**.405** (99.5)	.081 (99.5)	.000	.271
diabetes	**.661** (92.2)	.620 (92.9)	.487 (90.3)	.486 (87.0)	.600	.380	.278 (91.6)	.271 (90.3)	**.307** (90.9)	.281 (92.2)	**.307**	.239
ecoli	**.892** (100.)	.851 (100.)	.811 (100.)	.816 (98.5)	.381	.662	.466 (100.)	**.723** (100.)	.551 (100.)	.288 (100.)	.527	.587
fico	**.718** (93.4)	.674 (93.2)	.519 (92.9)	.342 (92.9)	.432	.371	**.346** (93.8)	.326 (92.1)	.234 (92.8)	.197 (93.8)	.108	.236
german	**.714** (87.0)	.629 (87.0)	.640 (86.0)	.541 (86.0)	.054	.315	.270 (87.0)	.209 (77.5)	**.403** (81.5)	.195 (83.5)	.160	.261
home	**.729** (93.9)	.671 (96.0)	.618 (93.9)	.613 (93.9)	.611	.355	**.506** (96.0)	.408 (96.0)	.452 (97.0)	.473 (94.9)	.407	.457
ionosphere	.678 (91.5)	**.741** (91.5)	.733 (87.3)	.479 (90.1)	.235	.216	**.718** (95.8)	.647 (94.4)	.456 (97.2)	.090 (97.2)	.086	.120
sonar	.723 (81.0)	**.756** (81.0)	.671 (76.2)	.673 (78.6)	.062	.325	.391 (85.7)	**.496** (85.7)	.259 (88.1)	.283 (92.9)	.138	.027
spam	.577 (96.5)	**.579** (96.9)	.425 (94.5)	.279 (94.6)	.351	.069	.234 (95.9)	**.238** (97.1)	.173 (94.6)	.046 (94.7)	.132	.232
titanic	**.830** (91.1)	.704 (91.1)	.682 (88.8)	.710 (89.9)	.433	.588	.621 (95.5)	.642 (95.0)	**.661** (97.2)	.594 (96.1)	.550	.541
wine	**.667** (66.3)	.629 (66.6)	.616 (60.4)	.563 (59.3)	.338	.485	.286 (71.0)	.285 (71.0)	.253 (70.6)	.262 (68.2)	.273	**.347**
yeast	**.644** (86.5)	.637 (88.2)	.503 (81.5)	.472 (81.8)	.314	.412	.385 (89.2)	.425 (89.9)	.434 (88.2)	**.448** (89.9)	.329	.436

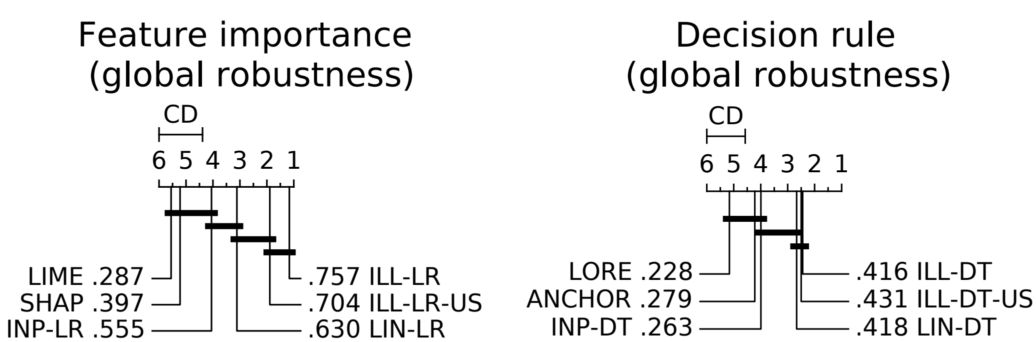

Fig. A8: Critical Difference plots with Nemenyi test at 90% confidence level for global robustness metrics in real-world datasets.